

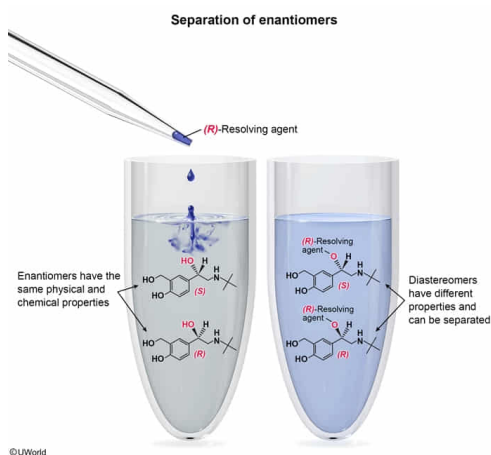
The (*R*)-enantiomer of the antiasthma drug albuterol is the active isomer. If a researcher wants to separate a racemic mixture of albuterol, which of the following methods will most likely separate the enantiomers?

- ☐ A. Extraction with dilute base [6%]
- ☐ B. Thin-layer chromatography [10%]
- ☐ C. Fractional distillation [24%]
- ✓ ☒ D. Addition of a resolving agent [58%]

Correct

58% answered correctly

Explanation:



An atom bonded to four unique substituents is called a **chiral center**. If corresponding chiral centers in a pair of otherwise identical molecules have the **opposite configuration** (ie, *R* in one molecule, *S* in another), the molecules are known as **enantiomers**, or nonsuperimposable mirror images. A **50:50 mixture** of enantiomers is known as a **racemic mixture**. Enantiomers have many of the same chemical and physical properties, including melting and boiling points, solubility, and polarity, although they differ in the way that they interact with **plane-polarized (linear) light**. Therefore, **separation** of enantiomers relies on **changing the physical properties** of the molecules.

The separation of enantiomers, such as those in the racemic mixture of albuterol, requires the addition of a **resolving agent** (a chiral molecule). When a resolving agent is added to a racemic mixture, it reacts with each enantiomer, forming a covalent bond or an ionic salt. Because the resolving agent is chiral, it incorporates a new chiral center into each enantiomer, creating a pair of **diastereomers**. Diastereomers can be separated from each other because, unlike enantiomers, they have different physical properties. Once the diastereomers are separated, **the resolving agent is removed**, yielding the original molecules as **single enantiomers**.

(Choices A, B, and C) Because enantiomers have the same chemical and physical properties, they cannot be separated by physical means such as extraction, thin-layer chromatography (TLC), or fractional distillation. Enantiomers react in the same way with bases, have the same retention time on a TLC plate, and have the same boiling point.

Educational objective:

A racemic mixture is a 50:50 mixture of enantiomers, which have the same chemical and physical properties, and therefore cannot be directly separated. Consequently, the separation of enantiomers requires the addition of a resolving agent to change their physical properties by creating a pair of diastereomers. The resolving agent is removed once the diastereomers are separated to yield the original molecules as single enantiomers.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401444

System:
Separation Techniques, Spectroscopy, and Analytical Methods